

Technical Stack Document

Campus Services Portal | Project implementation summary

Group A2 G1

Technology Overview

Area	Technology used
.NET version	.NET 8 (TargetFramework: net8.0); ASP.NET Core shared framework 8.x.
Frontend	Static HTML5 pages, CSS3, and vanilla JavaScript. The frontend is served from ASP.NET Core wwwroot.
Backend	ASP.NET Core Web API using MVC controllers, layered services and repositories, and dependency injection.
Database	Microsoft SQL Server, accessed through Entity Framework Core 8 and the SQL Server provider.
Programming languages	C# for backend and data layer; JavaScript for browser behavior; HTML and CSS for presentation; SQL Server SQL for persisted data.
APIs and services	RESTful API endpoints, JWT bearer authentication, Swagger/OpenAPI documentation, and SendGrid transactional email for verification and password reset.
Architecture	Controller -> Service -> Repository -> Entity Framework Core -> SQL Server. DTOs separate API contracts from database entities.

Libraries and Supporting Technologies

Package / tool	Version	Purpose
Microsoft.EntityFrameworkCore / SqlServer / Design / Tools	8.0.19	ORM, migrations, and SQL Server connectivity
Microsoft.AspNetCore.Authentication.JwtBearer	8.0.19	JWT authentication middleware
Microsoft.AspNetCore.OpenApi	8.0.19	OpenAPI endpoint support
Swashbuckle.AspNetCore	6.6.2	Swagger UI and OpenAPI generation
System.IdentityModel.Tokens.Jwt	8.22.0	JWT creation and token handling
BCrypt.Net-Next	4.2.0	Secure password hashing

Package / tool	Version	Purpose
SendGrid	9.29.3	Transactional email provider
MailKit and MimeKit	4.17.0	Installed email and MIME libraries

Implementation Notes

The project starts by applying Entity Framework Core migrations and inserting idempotent demonstration data. Static frontend files are hosted by the same ASP.NET Core application, avoiding a separate frontend build server. Development API documentation is available through Swagger UI.

Configuration values such as the SQL Server connection string, JWT signing key, and SendGrid API key are intentionally kept outside source control and supplied through local configuration or .NET user secrets.